

OpenBCI Meets the Web: A Scalable, Customizable Platform for Open-Source EEG Data Collection

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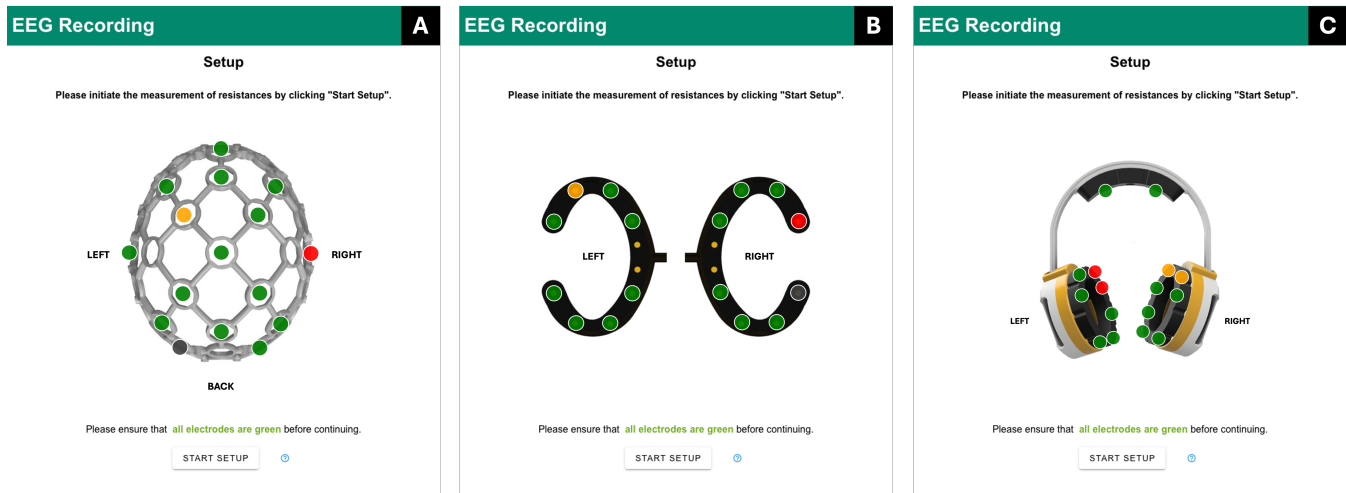


Figure 1: The Open-Source Web-App for Easy and Customizable Data Collection showing Impedance Checks with three Different Electrode Layout Configurations (A = OpenBCI Ultracortex; B = Open-cEEGrids, C = Open ExG Headphones)

Abstract

The traditional process of EEG-based BCI data collection is cumbersome and expertise-dependent due to complex hardware instrumentation, but also because data collection software typically comes as an on-premise solution with features tailored to trained experimenters. To overcome these limitations, we herein present a customizable platform for EEG data collection using OpenBCI hardware, that simplifies and accelerates the EEG setup process. The proposed system integrates various adaptations of the OpenBCI recording hardware, enabling easy customizability for versatile electrode configurations. The frontend, developed with Vue 3 and D3.js, provides real-time impedance checks and data visualization, while the backend, utilizing Apache, Node.js, and Flask, ensures efficient data transmission and storage. This platform addresses critical needs for cross-platform compatibility, customizability, efficient session management, and user-friendly interfaces, making it

ideal for large-scale and field studies. The automated data management and real-time feedback enhance data quality and reliability, supporting various research applications.

CCS Concepts

• **Human-centered computing** → *Empirical studies in HCI*; • **Hardware** → *Neural systems*.

Keywords

Wearable BCI, Headphones, EEG, ExG, OpenBCI, Open-cEEGrids

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1 Introduction

Throughout the last decade, HCI research has aimed to utilize brain-computer-interfaces (BCI) to revolutionize our interaction with technology, with passive applications like inferring sleep quality and monitoring mental states [2, 15, 23], as well as active systems for controlling digital and robotic devices [1, 25]. These BCI typically rely on electrical brain activity measured through an EEG and

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mostly occur in a laboratory setting, where a researcher fits the EEG device onto the participant's head, checks the impedance of the sensors, explains the experimental task, and monitors the incoming signals [7]. This process is highly labor- and expertise-intensive and creates a critical bottleneck for collecting larger datasets. Yet, in the BCI communities, awareness is increasing, that large data sets are necessary to extract more reliable information from the notoriously noisy EEG data, which exhibit significant intra- and inter-subject variability [14].

To overcome the scaling problem, researchers are leveraging dry-electrode wearable EEG systems, which are easier and faster to apply [16]. Numerous manufacturers compete in this field with proprietary hardware and software. In contrast, open-source projects like OpenBCI aim to democratize BCI research by enabling diverse groups to engage and customize their systems [18]. Examples include the Open-cEEGrids project [11] and the Open ExG Headphones [12], which adapt open-source tools into everyday BCI systems by making the recording hardware invisible.

Like other BCI tools, a critical bottleneck for larger studies with these devices remains in the on-premise recording software [7]. BCI software tools are typically used for setting up an EEG recording, including an initial system calibration phase (e.g., an impedance check), running a recording, and handling data storage or interfacing with custom software through APIs (e.g., LSL¹ or Brainflow²). These tools are designed for trained researchers and developers, not study participants. Recently, efforts have been made to create more participant-friendly, technology-agnostic platforms for BCI research through open-source mobile and web applications (e.g., MYND [9], SANS TRACAS [7]), focusing on the InteraXon Muse headband as the underlying EEG recording device.

Joining the rationale of democratizing BCI research through customizable open-source hardware and software, we believe similar development efforts are needed for large-scale data collections with OpenBCI hardware. There is a critical mismatch between custom hardware adaptations and the current OpenBCI GUI software, which lacks easy configurability (e.g., for new electrode positions) and only allows local data storage, requiring manual data collection. To address this, we are developing an open-source, user-friendly, and customizable solution for recording OpenBCI EEG data over internet protocols. This will enable scalable experiments and field studies with automated data streaming to secure institution servers, supervised by public research institutions. In this article, we present the first version of this recording platform, including descriptions of its architecture, workflow, and configurability. We also discuss remaining limitations and goals for the next versions.

2 Related Work

2.1 Wearable BCI Hardware

Recent advancements in wearable BCI technologies, particularly with the shift from stationary to wireless EEG, have been significant [16]. In HCI, some studies have focused on integrating these wearable EEG sensors into everyday items like glasses [13, 24], in-ear buds [15], around-the-ear gelled electrodes [6], and over-ear headphones [4]. These developments have also led to open-source

hardware adaptations like Open-cEEGrids [11] and Open ExG Headphones [12], based on OpenBCI's widely used open-source amplifiers. OpenBCI is the most widely adopted open-source BCI hardware project, with over 500 research publications³. However, while these previous works with the OpenBCI stack featured impressive advancements on the hardware side, researchers have either relied on the available on-premise software or on custom-built code interfaces to realize their studies [11, 12]. In other related open-source hardware projects, for example in the earables domain, projects like the esense [10] or the OpenEarable [20] can provide a reference point for the potential of wearable BCI, especially as they have seen more advanced developments of the underlying recording software aimed at large scale usability.

2.2 Wearable BCI Recording Software

The landscape of BCI recording software solutions can be split into code APIs, on-premise software, and web-app solutions, all of which aim to offer platform-agnostic recording solutions.

In the code API category, frameworks such as OpenViBE [19], BCI2000 [22], or Brainflow, are often used in research works to facilitate the development and operation of BCI systems with various EEG hardware. These frameworks primarily serve for building data collection and processing pipelines, which are beneficial for researchers and developers but often require substantial programming expertise.

In contrast, on-premise recording tools like the OpenBCI GUI⁴ offer a more plug-and-play-like functionality (e.g. for setting filters, checking signals, or visualizing frequency distributions), albeit with a steep learning curve geared towards EEG researchers. Therefore, a critical gap exists in the BCI software landscape regarding participant-friendly software for scaling up data collections. End-users need intuitive interfaces and minimal setup complexity, while researchers require tools that ensure robust data quality, automated data transfer security, and flexible configurations for different EEG hardware setups and electrode placements. Thus, software solutions that bridge this gap are essential.

The demand for such tools is underscored by related web applications designed for user-friendly data recording from devices, such as the Muse EEG headband (InteraXon – Toronto, Canada), in the MYND [9] or the SANS TRACAS projects [7]. Both platforms are still in their initial stages of development, and focus on different aspects, with MYND focusing on enabling smartphone recordings in field settings, and SANS TRACAS focusing on integrating behavioral experiments and EEG recordings in a web-based application. However, no such developments currently exist for the widely distributed OpenBCI hardware stack. Combined with recent advancements in web-based platforms for behavioral studies, like Pavlovica [7], such tools could significantly enhance the scalability of EEG recordings by enabling participant self-setup in labs or at home.

¹<https://github.com/scn/labstreaminglayer>

²<https://github.com/brainflow-dev/brainflow>

³<https://openbci.com/citations>

⁴https://github.com/OpenBCI/OpenBCI_GUI

3 System Description

3.1 Hardware Base

To bridge the gap between researchers and study participants for more large-scale adoption of wearable BCI technologies, we have developed a web application for the OpenBCI Cyton+Daisy amplifier system that features 16-channel ExG recordings with a sampling rate of up to 250 Hz, 24-bit resolution, low input-referred noise and low power consumption [18]. The Cyton and Daisy amplifiers can be separated, allowing for either 8- or 16-channel recordings, depending on the researchers' needs. In addition, the Cyton amplifier records 3-axis accelerometer data with a sampling rate of 25 Hz. Normally, this amplification system is used with a full-head harness, the Ultracortex MK IV (see Figure 1A), featuring the electrode positions of the standard 10-20 reference system [17] that are spread out over the head surface. Alternative configurations (mainly with different electrode positions) for the same amplifier can be found in the recent open-source projects for around-the-ear recordings [11, 12] - shown in Figures 1B and 1C.

3.2 Design Requirements

Our application's design enhances flexibility, customization, and usability for EEG research, making studies more efficient and scalable. We identified key design requirements by reviewing related literature [7, 9] and discussing with an interdisciplinary group of researchers before development.

3.2.1 Design Requirement 1 (DR1): Cross-platform availability. The application should be platform-independent, ensuring seamless operation across diverse hardware. Using maintainable, widely-used frameworks that support cross-platform compatibility will ensure flexibility and ease of maintenance. This guarantees broad accessibility, allowing researchers to deploy the application in various environments without compatibility issues.

3.2.2 Design Requirement 2 (DR2): Customizability. Researchers need diverse electrode setups to measure neurophysiological data accurately. Given the limited channels (8 or 16) in OpenBCI Cyton+Daisy hardware, electrode positions vary by study. Therefore, the web application must be easily customizable to accommodate different hardware configurations, optimizing data collection and enhancing flexibility and utility for various experiments.

3.2.3 Design Requirement 3 (DR3): Secure and efficient session management. As researchers scale experiments involving multiple participants, session handling becomes increasingly susceptible to errors. Therefore, there is a clear need for efficient session management to streamline operations, maintain data integrity and ensure that data is securely streamed to a provided server to prevent potential loss and protect sensitive data from breaches during transmission.

3.2.4 Design Requirement 4 (DR4): User-friendly interface. To scale the experiment effectively, enabling participants to configure the recording software themselves is crucial. Therefore, the setup process should be intuitive and require minimal user interaction, aiming for fewer components and less user input. Additionally, providing participants with immediate feedback on the duration of different steps and presenting information using easily interpretable

visuals is essential. The application should be a web application, so neither researchers in the laboratory nor participants during a field study at home need to install software on a local PC.

3.3 Solution Design

For the system architecture, we adopted a classic web application design comprising a frontend and several modular backend services, as illustrated in Figure 2. The web application was developed using robust JavaScript and Python frameworks to enhance efficiency and reliability, ensuring streamlined data handling and comprehensive visualization capabilities.

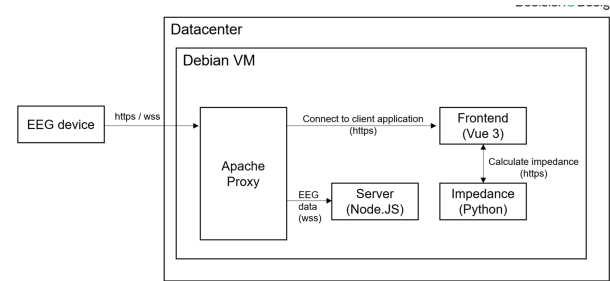


Figure 2: Architecture Overview

3.3.1 Solution for allowing cross-platform availability (DR1). To ensure platform independence, we developed the application using Vue 3 [26] and D3.js [3] for visualization. The backend uses Apache [8] and Node.js [5] for data management, while Flask [21] handles real-time impedance calculations. These cross-platform frameworks enable seamless operation across diverse hardware, maintaining flexibility and accessibility. This approach ensures easy maintenance and broad usability, allowing researchers to deploy the application without compatibility concerns.

3.3.2 Solution for customizability (DR2). The application enables researchers to tailor electrode setups to their specific neurophysiological data requirements using a JSON file for configuration, where each electrode is mapped to a channel of the EEG device. Additionally, the 3D model of the recording device can be dynamically adjusted, e.g. displaying headphones or Open-cEEGrids with their respective electrode layouts. Researchers have the flexibility to access these configurations via a URL parameter, thereby enhancing the adaptability of experimental designs.

3.3.3 Solution for secure and efficient session management (DR3). To manage data recordings efficiently, each participant can be assigned a unique ID, which also can be passed as a parameter in the URL. This feature ensures that the EEG data is systematically stored and easily retrievable based on participant ID. Because EEG data is streamed directly to a university server, participants and researchers are relieved from the task of manually collecting and managing files, thereby reducing the potential for manual errors. To ensure security, the data is transmitted via the WebSocket Secure (WSS) protocol, which encrypts it using SSL certificates. This encryption prevents the data from being intercepted and read by unauthorized parties, because the encrypted data cannot be easily deciphered without the appropriate decryption keys.

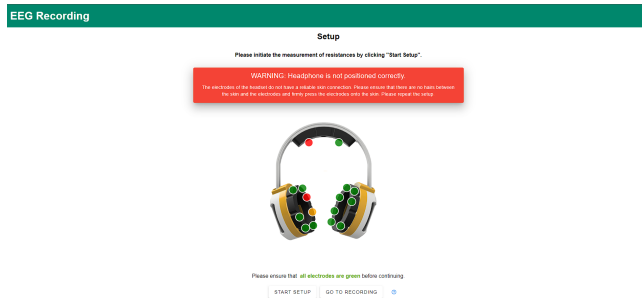


Figure 3: The user receives a notification that the impedance is not sufficient

3.3.4 Solution for user-friendly interface (DR4). The application features a user-friendly interface that automates impedance measurements upon headset connection, reducing user interaction. Impedance values before and after each session are stored on the server and displayed on a 3D model (see Figure 3). Electrodes are color-coded for connection quality: green (<750 kOhm), orange (750 - 1000 kOhm), and red (>1000 kOhm). Researchers can adjust these thresholds. This provides participants with real-time feedback to adjust electrodes for better impedance and aids researchers in comparing impedances for comprehensive data analysis.

3.4 Application Flow

Figure 4 illustrates the operational flow of our recording application. Section A provides an overview of its functionality, while Section B details each stage. To streamline user experience, participants connect the EEG headset via the Webserial API, triggering an automatic impedance check across all channels. This data is processed by a Flask microservice, segmented into 1-second windows, and filtered using a bandstop filter to remove 50Hz noise. A subsequent band-pass filter allows data between 27Hz and 37Hz to pass, isolating the 31.5Hz signal as configured in the OpenBCI GUI for consistency. Impedance results are visualized on a 3D model of the headset (Figure 3). If impedance levels are unsatisfactory, participants repeat the check until acceptable levels, set by researchers, are achieved. Impedance data is saved before and after each recording session. Once acceptable impedance is confirmed, EEG data recording begins, streamed in real time to a WebSocket server and stored by a Node.js server. At session end, recording stops, the WebSocket connection closes, and final impedance measurements are saved on the server for analysis.

4 Discussion and Conclusion

Our application focuses on developing an open-source, user-friendly, and highly customizable EEG recording solution tailored to meet diverse research needs. Primarily, this web based solution can reduce the need for researcher supervision and empower participants to set up recordings independently. The automated data management streamlines the organization and storage of EEG recordings, ensuring efficiency and data integrity throughout the research process. Real-time impedance tracking and visual feedback are integral features that improve the reliability and quality of data collection.

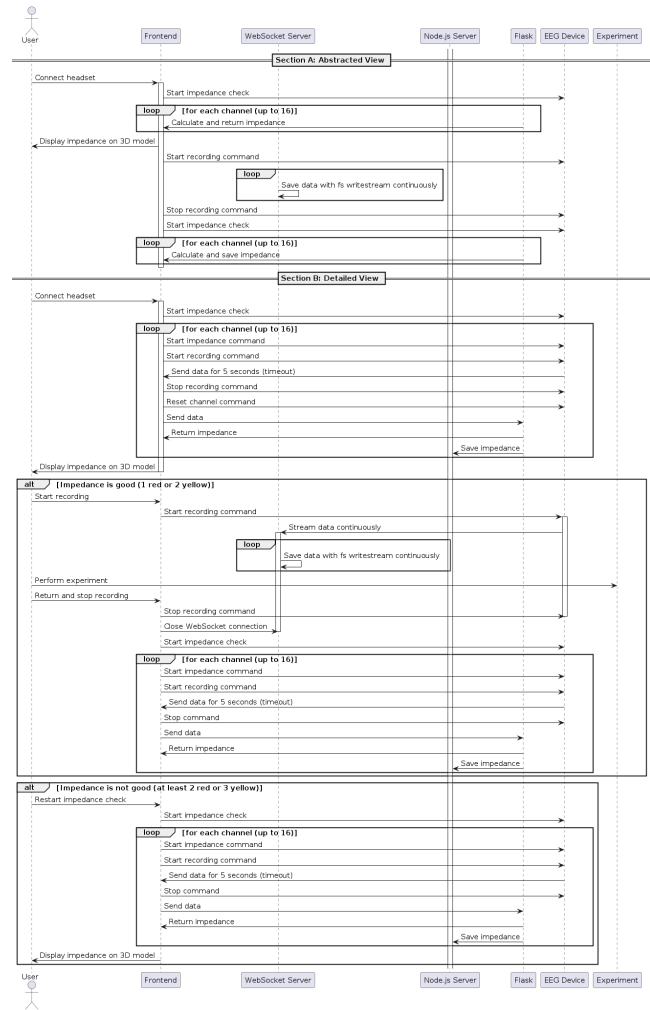


Figure 4: Application Flow

These capabilities not only support large-scale and field studies through remote and scalable data collection. We have already tested the setup with 20 participants, with 2-3 participants in the lab at the same time. However, continuous refinement based on user feedback is ongoing to optimize performance and usability, with future plans including the integration of preprocessing capabilities for EEG data and exploring collaborations with researchers and institutions to advance methodologies. Embracing open-source methodologies is crucial in broadening the scope of EEG studies, promoting accessibility, and enhancing the reliability of research findings through shared knowledge and resources.

5 Software availability

The codebase is available at GitHub: <https://github.com/Lukas-Stingl/eeg-collect>

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